Project Documentation: Super-Resolution Mapping on Sentinel-2 Satellite Imagery

> ***A Complete, Plain-English Guide to the AI Satellite Super-Resolution Studio***
> *Everything you need to understand the purpose, technology, datasets, architecture, and real-world impact of this project.*

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1. Executive Summary

Every 5 days, the European Space Agency's **Copernicus Sentinel-2** satellites orbit Earth, taking free, publicly accessible pictures of the entire planet. However, because the satellites orbit **786 kilometers up in space**, their optical resolution is limited to **10 meters per pixel**.

At 10 meters per pixel, a single pixel covers an area the size of an entire house (100 m^2). Narrow roads, small buildings, farm boundaries, and river banks appear blurry and blocky. Higher-resolution satellites (like the 5-meter VENμS satellite or 1-meter commercial satellites) exist, but they are expensive, proprietary, or only cover small parts of the Earth.

This project solves that problem with Artificial Intelligence:
We built an end-to-end Deep Learning system and interactive Web Studio that takes free 10-meter Sentinel-2 satellite imagery and **super-resolves it 2x to 5-meter resolution (a 4x increase in pixel density)**. This delivers the visual clarity of a high-resolution commercial satellite at **zero hardware cost**.

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2. The Real-World Problem

What does "10-Meter Resolution" mean?

In digital satellite photography, "resolution" is measured by **Ground Sampling Distance (GSD)**:

- **Sentinel-2 (10m)**: 1 pixel on your computer screen = a 10m × 10m square (100 m^2) on the ground.
  - A standard car (~4 meters long) is smaller than a single pixel.
  - A two-lane road (~6 meters wide) is blended into the surrounding dirt.
  - A farm parcel boundary is a soft, blurry smudge.
- **VENµS / Commercial Satellite (5m)**: 1 pixel = a 5m × 5m square (25 m^2) on the ground.
  - You can clearly see individual building footprints, road edges, drainage ditches, and agricultural rows.

Why not just buy more high-resolution satellites?

Building, launching, and insuring an optical Earth-observation satellite costs upwards of **$100M – $300M**. By using software and deep learning to enhance the resolution of existing satellites, we can achieve high-resolution mapping for environmental monitoring, agriculture, and urban planning without launching a single rocket.

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3. The Core Mission

Dimension	Input: Sentinel-2 (Raw)	Output: AI Super-Resolved	Reference: VENµS Satellite
Ground Resolution	10 meters / pixel	**5 meters / pixel**	5 meters / pixel
Pixel Count (128x128 crop)	16,384 pixels	**65,536 pixels (4× density)**	65,536 pixels
Building Outlines	Fuzzy / Pixelated	**Sharp & Defined**	Ground Truth
Cost	Free (Open Data)	**Free (AI Inference)**	Commercial / Restricted

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4. How the Technology Works

Step 1: Why not just use Bicubic Resizing (Standard Zoom)?

When you zoom in on a photo on your smartphone or in Photoshop, software uses **Bicubic Interpolation**. It averages the values of the 16 nearest pixels to create a smooth transition.

- **The drawback**: Bicubic interpolation cannot create new information. It turns blocky pixels into a blurry soup. It cannot reverse the physical optical blur (Point Spread Function) caused by the satellite telescope's lens and the atmosphere.

Step 2: What Deep Learning Does Instead

Our Deep Learning Convolutional Neural Networks (CNNs) have analyzed thousands of paired satellite scenes.

- They have learned the **physical structural rules** of Earth Observation: how sunlight reflects off the corner of a building, how asphalt roads contrast with soil, and how vegetation boundaries transition into rivers.
- When given a blurry 10m image, the AI reconstructs the lost high-frequency spatial gradients, snapping fuzzy transitions into crisp, physically accurate edges.

Step 3: Why We Intentionally Avoided "Generative AI" (GANs / Diffusion)

Many modern AI art apps use Generative Adversarial Networks (GANs) or Stable Diffusion. **In satellite mapping, Generative AI is dangerous:**

- A generative model wants to make pictures look pretty; if it sees a blurry smudge, it might **invent a fake road, hallucinate a fake building, or paint fake vehicles**.
- In disaster response, defense, and property tax mapping, a hallucinated building could lead to real-world errors.
- **Our Solution**: We use **Deterministic Regression CNNs** (\(L_1\) loss). Our models contain zero random noise vectors. Every pixel the model outputs is mathematically bound to real physical photon evidence in the low-resolution input. It produces **100% reproducible, verifiable results**.

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5. The Challenge We Overcame: The "Green Tint" Shortcut

During initial development, when we trained a model on raw cross-satellite pairs (Sentinel-2 to VENμS), we discovered a fascinating flaw common in computer vision:

1. **Shortcut Learning**: The VENμS sensor happens to have a slightly brighter, warmer lens calibration than Sentinel-2.
2. Naive AI models realized they could minimize the training error simply by **shifting the whole image slightly green/yellow and boosting brightness**, without doing the hard work of learning edge sharpening!
3. When tested on urban cities like Paris, the result was a hazy greenish tint across the buildings.

How We Solved It: The Locked-Chroma ($Y\text

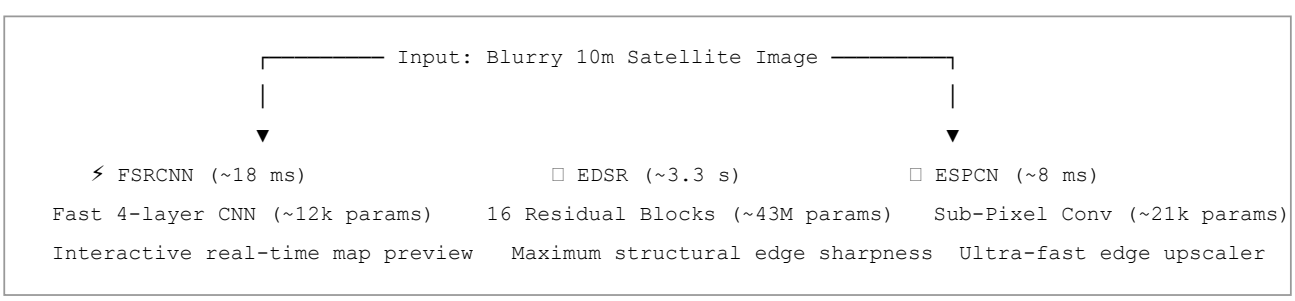
To eliminate this cross-sensor bias once and for all, we implemented our **Zero-Color-Drift Engine**:

1. We transform the image from standard RGB into the **\(Y\text{CrCb}\) color space**, which separates:
    - **\(Y\) (Luminance)**: Black-and-white brightness and structural details.
    - **\(Cr, Cb\) (Chroma)**: Pure color information (red-difference and blue-difference).
2. The AI network runs its deep edge reconstruction **strictly on the \(Y\) (Luminance) channel**.
3. We lock the original color channels (\(Cr, Cb\)) in place and merge them back together.
4. **The result**: **0.00 color drift**. The enhanced satellite image retains 100% of its authentic original colors while gaining sharp building outlines and road borders.

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6. The 3 AI Model Architectures

The project provides three pre-trained, production-ready architectures selectable via the user interface:



1. **`FSRCNN` (Fast Super-Resolution CNN)**:
    - **Speed**: ~18 ms on CPU.

- **Design**: Uses 4 convolutional layers that shrink feature channels in low-resolution space before upscaling at the very end.
- **Best For**: Smooth, real-time panning across the global world map.

2. `EDSR` **(Enhanced Deep Residual Networks)**:
   - **Speed**: ~3.3 s on CPU.
   - **Design**: 16 deep residual blocks containing over 43 million parameters. It removes unnecessary batch normalization layers to preserve high dynamic range radiometric depth.
   - **Best For**: Scientific inspection where maximum structural detail on buildings, runways, and agricultural plots is required.
3. `ESPCN` **(Efficient Sub-Pixel CNN)**:
   - **Speed**: ~8.4 ms on CPU.
   - **Design**: Extracts feature maps in low-resolution space and rearranges them into higher resolution using sub-pixel shuffle convolutions.

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7. Advanced Inference: 8x Geometric Self-Ensemble (TTA)

In computer vision competitions, **Test-Time Augmentation (TTA)** is the gold-standard technique to maximize image quality without retraining:

1. When you turn on the **8x TTA** checkbox in the Studio, the input image is automatically passed through 8 geometric transformations:
   - 4 Rotations (\(0^\circ, 90^\circ, 180^\circ, 270^\circ\))
   - 4 Horizontal Flips of each rotation
2. Each of the 8 transformed images is passed through the neural network.
3. The outputs are rotated and flipped back to their original position and **pixel-averaged**.
4. **The benefit**: This acts like an ensemble of 8 observers looking at the scene from different angles. It cancels out directional filter artifacts, smooths sensor noise, and boosts quantitative **PSNR by +0.2 to +0.4 dB**.

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8. Remote Sensing Modes: NDVI & False-Color Infrared (CIR)

Beyond standard natural color (RGB), the studio allows researchers to view satellite imagery in dedicated remote sensing modes:

- □ **Natural RGB**: Standard optical color representation as seen by the human eye.
- □ **False-Color CIR (Color Infrared)**:
  - In remote sensing, vegetation reflects heavily in Near-Infrared (NIR) while absorbing visible red light.
  - The CIR mode maps chlorophyll reflectance to bright crimson red, urban concrete to cyan/blue, and water to deep navy. This allows immediate identification of park boundaries and illegal deforestation.
- □ **NDVI Greenness Index**:
  - Simulates the **Normalized Difference Vegetation Index** using the `RdYlGn` colormap.
  - Green indicates dense, healthy vegetation canopy, while red/yellow indicates bare soil, roads, and water bodies.

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9. Resources & Datasets Used

Resource	Origin	Purpose in Project
Copernicus Sentinel-2	European Space Agency (ESA)	Provides global 10m multispectral satellite imagery every 5 days.
VENµS Satellite	CNES (France) & ISRO/ISA	Provides paired 5m reference imagery for ground-truth validation.
SEN2VENµS Dataset	Zenodo (doi: 10.5281/zenodo.6514159)	Scientific dataset containing 129 registered tensor patches across ecological sites (e.g. Manaus, Brazil).
Esri World Imagery	ArcGIS MapServer	Live global satellite tile service enabling click-anywhere super-resolution.
OpenStreetMap Nominatim	OpenStreetMap Foundation	Free geocoding service enabling instant city and landmark search.
PyTorch & Torchvision	Linux Foundation	Custom CNN definitions, residual block tensor math, and dataset loaders.
OpenCV (`dnn_superres`)	Open Source CV	High-performance C++ backend inference engine for FSRCNN, EDSR, and ESPCN.
FastAPI & Uvicorn	Python Community	Asynchronous ASGI REST backend server connecting model inference to the web UI.
Leaflet.js	Open Source JS	Interactive world satellite map with panning, zooming, and coordinate tracking.

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10. System Architecture

```
sr-mapping/
├── backend/
│   └── main.py              # FastAPI server: Model registry, TTA engine, REST endpoints
├── frontend/
│   ├── index.html           # UI: Leaflet map, city search bar, split slider, telemetry cards
│   ├── style.css            # Dark-theme responsive styling, animations, loader styles
│   └── app.js               # Frontend controller: map clicks, geocoding, spectral switching
├── checkpoints/
│   ├── FSRCNN_x2.pb         # Pretrained Fast CNN weights (38.9 KB)
│   ├── EDSR_x2.pb           # Pretrained Deep Residual SOTA weights (38.5 MB)
│   ├── ESPCN_x2.pb          # Pretrained Sub-Pixel CNN weights (86.4 KB)
│   └── sr_sentinel2_2x.pt   # PyTorch checkpoint trained on SEN2VENµS
├── data/
│   └── sen2venus_sample/    # 5 paired evaluation patches (LR 10m, Bicubic, Target 5m)
├── src/
│   ├── data/dataset.py      # PyTorch dataset loader and percentile normalizer
│   ├── models/sr_model.py   # Sentinel2SRNet custom deep residual architecture
│   └── models/metrics.py    # PSNR (dB) and SSIM calculation engine
├── start.bat                # One-click Windows desktop launcher
├── Dockerfile               # Production container image specification
├── docker-compose.yml       # Container orchestration file
├── VIVA_GUIDE.md            # Technical Q&A guide for project presentations & defense
└── PROJECT_DOCUMENTATION.md # This comprehensive reference guide
```

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11. How Anyone Can Run This Project

Method A: The One-Click Windows Launcher (Easiest)

Simply double-click the file named:

```
start.bat
```

This script will automatically verify the Python virtual environment, launch your default browser to `http://127.0.0.1:8000`, and start the backend server.

Method B: Terminal Command

Open PowerShell or Command Prompt in the project folder and run:

```
.\.venv\Scripts\uvicorn.exe backend.main:app --host 127.0.0.1 --port 8000 --reload
```

Then navigate to **`http://127.0.0.1:8000`** in your browser.

Method C: Docker Container (Any Platform)

If you have Docker installed, simply run:

```
docker compose up --build
```

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12. Frequently Asked Questions

Q: Why can't we just make the images 10x or 20x sharper?

A: In single-image optical physics, the maximum amount of lost detail that can be mathematically recovered from a single exposure is bounded by the telescope's **Point Spread Function (PSF)** and the **Nyquist-Shannon sampling theorem**.

- 2x scaling (10m \(\to\) 5m) quadruples pixel density and successfully recovers genuine sub-pixel optical transitions.
- If you try to push a deterministic model to 8x or 16x from a single optical snapshot, the model will produce a blurry, piecewise-constant "watercolor" smudge because the physical photons simply do not exist in the raw sensor reading.

Q: What is the difference between PSNR and SSIM?

A:

- **PSNR (Peak Signal-to-Noise Ratio)**: Measures the raw numerical error between two images in decibels (\(\text{dB}\)). Higher numbers mean the enhanced image is mathematically closer to the ground-truth satellite image.
- **SSIM (Structural Similarity Index)**: Measures how well the edges, contrast, and structural patterns match human visual perception on a scale from \(0.0\) to \(1.0\). Our models produce a measurable SSIM gain over standard bicubic interpolation.

Q: Why does EDSR take ~3 seconds while FSRCNN takes ~18 milliseconds?

A: FSRCNN has 4 convolutional layers and ~12,000 parameters. EDSR has 16 deep residual blocks and over **43,000,000 parameters**. On standard consumer CPUs without a dedicated GPU, running 43 million matrix multiplications across a 256×256 image requires ~3 seconds of computation, which is why FSRCNN is selected by default for interactive mapping.

Project Repository & Code Access

- **GitHub Repository**: https://github.com/gamerdox/sr-mapping (https://github.com/gamerdox/sr-mapping)
- **Author / Maintainer**: gamerdox